# Design notes

DIY CNC epoxy-granite mill · October 2026

Fixed-gantry, epoxy-granite desktop mill sized like the Shariff DMC2 Mini (180 x 300 x 140 mm travel), with a 2020/ACM enclosure mounted on the base and a clear lift-up door.

This document explains the design: the frame, the coolant system, the stiffness study and the cost. The repository layout and the rebuild commands are in the top-level `README.md`.

## Frame

Five epoxy-granite castings that bolt together, each with cast-in steel:

| Module | Steel inside | Joins to |
|---|---|---|
| Base front | 2x 2x2x1/8 in tubes (12 in) under the table rails, hollow, sockets for the split spigots, 2x 1/2-13 draw rods, the coolant outlet | Rear half: 1-1/2 in spigots slide inside the sockets (printed sleeves take up the clearance); the draw rods pull it tight from pockets in the front face. Cast against the rear half, so the joint fits by construction |
| Base rear | 2x 2x2x1/8 in tubes (12 in), 2x 1-1/2 in spigots (6 in), draw-rod coupling nuts, the uprights' dowels and anchor bars, O-ring groove on the split face | - |
| Upright (x2), a raked buttress | A two-layer 1 in steel top plate, O-ring groove under the foot | Two dowels in the base locate it (grouted into oversize holes in the foot at first assembly), 4x 1/2-13 tie rods pull it down: two from nuts held in the top plate, two from pockets in the sloped front, all right through the base to 3/8 x 3 in steel anchor bars in its underside |
| Beam (140 deep x 140 high) | 2x 2x2x1/8 in tubes (18 in), drilled 3/4 in where the inner cap screws pass | A dowel (grouted into an oversize plate hole at first assembly) and 4x 3/8-16 bolts per side into the upright top plates |

**Nothing you buy gets cut.** The rails (including the 600 mm Y pair), ballscrews and tube are used at the lengths they're sold in: tube in 18, 12 and 6 in pieces of two US sizes (remnants are fine, lengths +/- 1/4 in, 0.120 in wall works as well as 1/8 in). The risers, aluminum plates are ordered cut to size, and the upright top plates and the base anchor bars are laser cut from the DXFs in `docs/1-sourcing/laser/`. The only cutting left is the 1/2-13 threaded rod. The only machining is drilling the beam tubes (4 holes each), drilling and tapping the risers and support pads, and tapping the top plates and anchor bars. Use plain steel, not galvanized: epoxy bonds poorly to zinc. Wire-brush off rust and loose scale and degrease just before casting; paint only the part of each leg that hangs below the base.

**Keeping the epoxy down.** About 10.3 kg of epoxy (8.5 % of the 121 kg the castings would weigh with sand; 130 kg with the alumina):

- The base tubes and legs stay hollow. Printed plugs close their ends so no mix gets in; fill them with dry sand first if you want extra damping. The beam tubes are filled, because the cap screws pass through them.
- The base's underside pockets follow the stiffness study below: two full-depth girders (80 mm wide, the narrowest that still fills around the rail tubes) under the table rails, a full-depth tie between them across the split, a full-depth rib across the front of each upright foot and another along the split, and a 20 mm top skin almost everywhere else, including over pockets under the upright feet that keep clear of the band under each upright where the tie rods run

down to the anchor bar. The 70 mm front wall carries the coolant trough. Every pocket is open to the bottom, so the cores (printed shells or foam blocks) pull straight out; a few are shallow steps (40 or 65 mm deep) under the trough band, the draw rods, the knees and outboard of the anchor bars (`base_pockets` in `model/build_cnc.py`).

**Molds and fits.** The molds are a kit (`model/moldbox.py`, the mold guide): melamine boards with every wall 25 mm above the pour line, held by printed brackets and saddles with M4 bolts through that band, so no bolt passes through a cast face. Every printed piece bolts to its board from outside into a captive nut (a bayonet pocket in the face against the board), or pins on if it is too thin, and pieces cut to fit the printer are pegged together. The hidden pocket cores are XPS foam, which keeps the printing to 1.4 kg of PETG (sliced in Bambu Studio). A clash check in the model keeps the kit, the boards, the castings and the steel apart. Two joints are made to fit by construction instead of by measuring: the front base half is cast against the rear half (flush top faces, matching split faces, tubes straight onto the spigots), and the dowels are grouted into oversize holes at first assembly, so two independent molds don't have to agree.

- The mix matters as much as the geometry. Use a well-graded aggregate (gravel up to about 10 mm, sand, fine sand, and whiting, i.e. calcium carbonate powder) and vibrate it. The split in the build guide follows a modified Fuller curve, P(d) = (d^q - d_min^q) / (D^q - d_min^q) with D = 10 mm and q about 0.45: q = 0.5 (classic Fuller) is a little coarse and needs more epoxy to wet, q = 0.37 (Andreasen) packs tighter but its extra powder soaks up epoxy.
- **Aluminum oxide in the beam and uprights.** Alumina grit (E about 370 GPa for brown fused, a little less for black, against about 70 for quartz and granite) replaces the sand, volume for volume, in those three castings only (`analysis/alumina.py`): brown #16 for the coarse sand, black 70 grit for the fine sand. A differential-effective-medium estimate of the mix (the plain mix comes out at 34 GPa, close to the 30 GPa the frame model uses) puts it 22 % stiffer with the #16 alone, 10 % with the 70 grit alone and 37 % with both. On the frame, both come to about +17 / +16 / +21 % in X / Y / Z for $150 (a 50 lb and a 25 lb bag); the #16 alone is about +10 %. Doing the base as well would take about 60 kg more grit for another 8-11 %. It's an estimate, so the build guide's tap-tested bars check it, and sand is the fallback. Other options: **Graded industrial quartz** (Agsco and similar) is clean, dry and consistent but costs more and ships heavy; pool filter sand is the local equivalent. **Ceramic microspheres** (Zeeospheres) help the mix flow at low epoxy, but whiting does most of that for far less. **Epoxy grout** (Laticrete SpectraLock) is fine for a test bar or a repair, but it's fine-grained (about 15-20 % binder) and far too expensive for 130 kg. Whiting is used instead of silica flour because it has no crystalline silica: a plain dust mask is enough. A good mix gets to 8.5 % epoxy, which fits a single 3 gal kit (10 % needs a second, small kit); the build guide's jar test and test bars check it before anything is cast.

**What the rails and supports bolt to.** Nothing bolts to the cast-in tubes; they stiffen the castings and form the joints. The rails screw into steel risers, and the risers are bedded on the castings with a thin epoxy grout and held down by counterbored M5 screws (under the rail, between its own screws) into cast-in inserts: 5 per riser on the base (Y), 4 on the beam (X). The Y ballscrew supports (BK12/BF12) bolt into M6 inserts in the base top. The X supports (BK10/BF10) sit over the front beam tube, where an insert won't fit, so they bolt to two 1/4 in steel pads cast flush into the beam face and bedded onto the tube with thickened epoxy. The Z axis mounts on the aluminum plates and needs no inserts. See the build guide.

**Load path.** The castings carry the load; the steel is where load changes hands. A cutting force goes spindle, Z plate, Z and X blocks, X rails, steel risers, beam casting; then at each end four 3/8-16 grade 8 bolts and a dowel into a 3/8 in steel plate cast into the upright top; the upright casting; four 1/2-13 tie rods per upright, from nuts held in the top plate (or in the slope pockets) right through the upright and the base to a 3/8 x 3 in steel anchor bar in the base underside, nut and washer below it, with the foot bearing on the base top, friction under that clamp taking the side load and two grouted dowels locating it; the base halves, clamped face to face by two draw rods, with the steel spigots in the rail tubes keeping them in line; and back up through the Y risers, rails and blocks to the table. Every bolted joint is steel thread in steel (plates, couplers, nuts) or a cast-in brass insert, never a thread cut in epoxy granite,

and every joint bears on a big flat epoxy face under preload. The uprights in particular hang on one steel chain: the beam screws pull up on the top plate, the tie rods pull it down, and the anchor bars take the rods under the base, so the epoxy granite between them only ever sees compression, and nothing that holds an upright depends on a thread or a bond in the casting. All eight tie-rod nuts are under the base or in the slope pockets, where they can be checked and retightened. The stiffness study treats the joints as bonded, which is what that preload is for.

The steel is tied together with bolts, not welded. A welded skeleton would have to be one casting (about 150 kg, one mold) or be welded after casting (the heat ruins the epoxy next to it), and it would make the frame one piece that can't be moved in parts, re-cast one member at a time, or rebuilt with a longer beam. Bolted, the chain from the beam to the anchor bars is just as continuous in tension, and it comes apart. The tubes inside the members aren't part of it; the study found them worth 1 % (base) and 6-7 % (beam), and the frame is already about 2-4 times stiffer at the tool (84-125 N/um) than the ballscrews (30-50 N/um). What does pay is keeping the joints flat, clean and preloaded, which is why they're cast against each other or grouted.

**Steel tolerances.** Nothing is cut at home except the threaded rod, but a supplier's saw cut is only good to about 1/8 in, and A500 tube is about 0.5 mm off nominal with a weld seam inside. Each cast-in piece is set from one end, and its other end has room (`tube_cut_tol`, checked in the report). The base tubes are set from the split end and the legs from their holders; the beam tubes are set by the rods through their 3/4 in holes, marked from each tube's middle, and their ends float on the locators; the tie rods are screwed up flush at the top and have a 6 mm tail under the bottom nut. Printed parts that fit on or into tube have 0.5 mm a side (`tube_fit`) and grooves for the seam. The X support pads are bedded onto the beam tube with thickened epoxy across a 1.5 mm gap instead of being expected to touch it. If the tube that arrives is further off, put its calipered sizes in `frame_tube` / `post_tube` and re-run before printing.

The enclosure's 2020 frame bolts to the base top with angle brackets into cast-in M5 inserts, so the base is the enclosure floor. Seal the top with an epoxy gel coat; coolant runs to the two drains at the front. The door laps over the base front face.

Assembly: base halves (spigots in, tighten the draw rods), uprights (dowels grouted into the feet the first time, tighten the tie rods), beam (cap screws), axes, enclosure. To break it down, take the X and Y motors off first (they pass through the enclosure panels), then reverse the order. Heaviest piece: the rear base half (about 42 kg). The upright feet run forward across the base split, so the base halves have to be joined before the uprights go on.

## Stiffness study (`analysis/`)

A finite-element model of the frame was used to see where the epoxy granite does the most good, and a topology optimization was run on the base.

**Model.** The castings and cast-in steel are voxelized from this model at 10 mm (66k brick elements). Each voxel's stiffness is its mix of epoxy granite (E = 30 GPa) and steel; the joints between castings are treated as bonded, which is what the preload is for. A 1 kN force on the tool goes into the beam through the X blocks (the X component through the X ballscrew to its BK bearing) and, reversed, into the base through the Y blocks (the Y component through the Y screw to its BK). The result is the frame's own stiffness at the tool, tool against work, for X at the center and end, the table at both ends and the center, and the tool 50 mm above the table. Rails, blocks, screws and the spindle are not in it; the screws alone are roughly 30-50 N/um, so the frame should stay well above that.

**Findings** (frame only, N/um at the tool, mean over the load cases):

| Frame | Epoxy granite | X | Y | Z |
|---|---|---|---|---|
| Previous design | 42.4 L | 48.9 | 80.7 | 57.9 |
| Same, base made rigid | | 87.2 | 130.9 | 80.2 |
| Same, uprights rigid | | 83.3 | 103.9 | 102.6 |
| Same, beam rigid | | 122.9 | 175.7 | 125.9 |
| Base with no pockets at all (+14 L) | 56.8 L | 61.5 | 90.1 | 67.0 |
| Beam 120 -> 140 mm deep only (+1.5 L) | 43.9 L | 51.1 | 90.9 | 63.8 |
| Optimized base + 140 mm beam | 43.7 L | 54.0 | 94.5 | 68.0 |
| + riser hold-downs, support pads, uncut 600 mm Y rails | 43.7 L | 55.7 | 96.8 | 68.9 |
| Base + uprights optimized together, same material (free-form uprights) | 43.7 L | 63.9 | 103.3 | 85.2 |
| Raked-buttress uprights, base re-laid around them | 52.6 L | 77.5 | 107.5 | 93.0 |
| + blunt upright toe, rounded edges, deeper slope pockets | 53.0 L | 78.2 | 110.1 | 95.5 |
| + alumina grit in the beam and uprights (estimate) | 53.0 L | 91.3 | 127.3 | 115.7 |
| Lighter base (pockets under the upright feet, 20 mm skin), 9 kg less | 49.5 L | 84.3 | 124.5 | 111.1 |
| Same, tie rods not counted as glued in (they sit in clearance holes; the rows below do the same; `vox_before_chain`) | 49.3 L | 81.6 | 121.9 | 107.6 |
| + tie rods right through the base to steel anchor bars, two-layer top plates | 50.5 L | 86.8 | 124.7 | 111.6 |
| + posts and sockets taken out, a second dowel per foot | 52.4 L | 86.3 | 124.2 | 110.1 |
| **Now: + upright feet 260 mm ahead of the beam (from 290)** | **51.3 L** | **84.2** | **123.0** | **109.3** |

- **The steel chain** (tie rods from the top plates to anchor bars under the base) is worth +6 / +2 / +4 % (the bars, the thicker top plates and the solid band left under each foot together). It also makes the posts redundant: with the feet clamped through steel they were worth under 1.5 %, so they're gone, and with them the sockets, the post bushings and the post grouting (a second grouted dowel per foot locates it instead). The stiffness left over paid for 30 mm off each upright foot. The frame ends up +3 / +1 / +2 % stiffer than before the chain, with modes 136 / 268 / 303 Hz (from 132 / 265 / 282), and about 6 kg heavier (the bars and the thicker top plates). A 250 mm foot keeps the static stiffness but drops mode 2 below the old frame; shorter than that the front foot pockets run past their mold bridges. Thinner uprights and a thinner front wall didn't pay.
- The beam is the biggest share of the deflection, then the base and the uprights about equally. Liter for liter, epoxy in the beam is worth roughly ten times as much as epoxy in the base, so the beam now fills the upright tops (140 deep instead of 120).
- In the new frame the beam is still the biggest share. Making the uprights rigid would add about 75 / 25 / 85 % in X / Y / Z, and making the base rigid about 70 / 70 / 35 %. The uprights are only 3.1 L each, so per liter they're the next place to add material (wider or deeper, which also grows the base and the enclosure).
- **Uprights.** Optimizing the base and the uprights together (`analysis/topopt_frame.py`, one shared budget, the uprights free to grow anywhere beside the table under the beam) moves about 3 L from the base into the uprights and turns each upright into a raked buttress: the foot runs about 300 mm forward along the base and the front face slopes up to just ahead of the beam (`analysis/figs/frame_34.6_upright.png`). That alone, at the same total material, is +15 / +7 / +24 % in X / Y / Z. With 4 L less material in total the frame ends up about 20 % less stiff than before, so cutting material isn't worth it.

- The castable version drawn now is a solid full-width buttress (one sloped board in the mold, no cores) with its foot 290 mm ahead of the beam face: 5.3 L per upright (5.5 L since the toe got a 20 mm nose). The base was then re-optimized around the longer feet (`analysis/figs/frame2_base_*.png`): it wants full depth under the feet and a wider tie across the split. Total +8.9 L of epoxy granite (about $75) for +39 / +11 / +35 %.
- The base optimization (`analysis/topopt.py`, pockets open to the bottom) puts the base material into girders under the rails, a tie between them behind the split and knees to the sockets, and takes it out of the perimeter walls, the cross rib at the split and the skin. The castable version drawn here has the same volume as before (28.4 L) and is 5 % stiffer on average (at least 3 % in every load case); the beam change adds the rest.
- The cast-in steel is mostly there for the joints. Replacing the base tubes with solid casting would cost only about 1 %; the beam tubes are worth 6-7 %. Six #6 rebars instead of the beam tubes would be about 3 % less stiff, so the tubes stay.
- Cutting epoxy from the base is cheap in money but expensive in stiffness. At the optimized layout's best: 24.5 L matches the old base, 21 L loses 11 %, and below about 18.5 L (the minimum skin, walls and cover around the steel) it falls off fast. Epoxy granite costs about $8.50/L, so even the best case saves about $35. A free-form base with closed box sections (lost foam cores) is only about 8 % stiffer than open pockets at the same volume, which isn't worth the trouble.
- **Weight** (`analysis/weight.py`, `to_light_*`). With the alumina grit in the beam and uprights there was stiffness to spare, so the base was optimized again at smaller budgets. Free-form (ideal) pockets: 27 L of base is 14 kg lighter for -7 / -1 / -2 %, 24 L is 21 kg lighter and still at the sand-only frame's stiffness, 21 L drops below it. Nearly all of it comes out from under the upright feet, which the earlier, larger budget had filled. The castable version keeps a rib across the front of each foot and along the split and leaves room around the tie-rod couplers and the sockets: base 32.8 -> 29.1 L, castings 9 kg lighter, frame -8 / -2 / -4 % against the alumina frame and still +8 / +13 / +16 % over the sand-only one. The simple trims don't pay: 60 mm girders leave under 5 mm of casting beside the rail tubes, and a 55 mm front wall runs the pockets into the legs.

**Material and the 8.5 % mix.** The model uses E = 30 GPa for the epoxy granite. Published values for well-graded 7-10 % mixes are 25-45 GPa (one commercial mineral casting lists 25 GPa); the university studies use 15-30 % resin with powder fillers and aren't comparable. Granite is about 70 GPa and the epoxy about 3 GPa, so a lower epoxy fraction is as stiff or stiffer as long as every stone is wetted and there are no voids; what it gives up is a little damping (the resin does the damping) and some margin against dry spots. The frame's stiffness scales almost directly with E:

| E of the epoxy granite | X | Y | Z |
|---|---|---|---|
| 20 GPa | 58.0 | 80.3 | 72.3 |
| 25 GPa | 68.3 | 95.5 | 84.1 |
| **30 GPa (assumed)** | **78.2** | **110.1** | **95.5** |
| 35 GPa | 87.9 | 124.4 | 106.6 |
| 40 GPa | 97.3 | 138.4 | 117.4 |

So the packing of the mix matters more than the last percent of epoxy. The build guide's jar test and tap-tested bars check both before anything is cast.

**Vibration** (`analysis/modal.py`, run with `FEA_GRIT=1.37` for the alumina). The frame as drawn on its four feet, with 12 kg of X carriage, Z axis and spindle on the beam blocks and 12 kg of table, vise and work on the Y blocks:

| Mode | Frequency | Motion |
|---|---|---|
| 1 | 136 Hz | gantry sways side to side (X); base twisting on its feet carries 42 % of it |

| Mode | Frequency | Motion |
|---|---|---|
| 2 | 268 Hz | beam rocks front to back (Y) |
| 3 | 303 Hz | base bends, front to back |
| 4 | 362 Hz | gantry sway with the base, second form |

Bricks this coarse run a little stiff, so the real frequencies are probably 10-20 % lower. Mode 1 is at the bottom of the tooth-pass band for steel (roughly 130-400 Hz: 4,000-8,000 rpm with 2-3 flutes) and modes 2-4 are inside it. That's normal for a machine this size; the frame is still several times stiffer than the spindle and tool (static X/Y/Z above), and chatter on a mill like this usually comes from the spindle, Z axis and tool. If a frequency rings, change the spindle speed by 10-20 % rather than the frame. A solid bench (the base twisting on its feet is in every mode) and sand in the hollow tubes and legs help more than geometry changes.

**Lightening holes in the uprights** (`analysis/topopt_upright.py`, `compare_uprights.py`). The uprights were optimized again on their own, with the cuts forced to go straight through (windows) or to stop at a 20 mm web on the inner face (pockets from the outside), keeping 90 % and 80 % of their material (run on the frame before the toe and edge cleanup, which added 1-3 %):

| Uprights (both) | Removed | X | Y | Z | Mode 1 / 2 | Epoxy saved |
|---|---|---|---|---|---|---|
| Solid, as drawn | | 77.5 | 107.5 | 93.0 | 135 / 264 Hz | |
| One window each, 90 % | 1.05 L, 2.5 kg | 75.0 | 106.5 | 91.2 | 136 / 263 Hz | ~$3 |
| One outer pocket each, 90 % | 1.05 L | 75.6 | 106.8 | 92.0 | 136 / 261 Hz | ~$3 |
| Two windows each, 80 % | 2.1 L, 4.9 kg | 71.7 | 104.8 | 88.9 | 136 / 255 Hz | ~$6 |
| Two outer pockets each, 80 % | 2.1 L | 73.1 | 105.6 | 90.3 | 135 / 254 Hz | ~$6 |

The holes land where you'd expect, in the low-stress middle of the buttress just ahead of the front post (`analysis/figs/upright_*.png`), but they don't pay: 10 % of the upright costs 1-3 % of the frame's stiffness and saves about $3 of epoxy (the 3 gal kit is bought either way), and the natural frequencies don't move because the mass that matters is the beam and the head on top. Each window would also need a printed liner. The uprights stay solid; the frame optimization above already moved material into them, not out.

Figures in `analysis/figs` (and `renders/frame_base_underside.png`): `frame_34.6_*` (base and uprights optimized together), `frame2_base_*` (base around the new uprights), `base_current.png` (the previous base), `base_final.png` (solid depth from the top in each column), `base_pocket_*` and `base_free_*` (optimization results at 18-28.5 L), and `tradeoff.png` (base stiffness against base volume), `upright_*` (lightening holes, at 90 and 80 % of the upright).

Rerun it with a Python that has numpy, scipy, matplotlib and scikit-sparse 0.5 or later (older scikit-sparse falls back to a much slower solver):

```
/Applications/FreeCAD.app/Contents/Resources/bin/freecadcmd analysis/voxelize.py   # ->
analysis/vox_current.*
cd analysis
python baseline.py                                  # stiffness and each module's share (about 1
min)
FEA_VOX=vox_previous sh run_sweep.sh                # base optimizations (about 10 min each)
FEA_VOX=vox_previous python evaluate.py to_*.npz    # table + figures
FEA_VOX=vox_current python whatif.py                # stiffness of the model as drawn
```

```
# voxelize with VOX_SKIP=TieRod,DrawRod so the rods in their clearance holes don't count as glued
in
python alumina.py                                    # alumina grit: mix modulus and frame stiffness
FEA_GRIT=1.37 python -c "import topopt; topopt.main('pocket', 24, tag='light_24')"   # lighter base
python weight.py vox_current [vox_...]               # castings mass against stiffness
python modal.py                                      # natural frequencies (about 30 s)
python topopt_upright.py 0.9 [web]                   # upright holes (about 10 min each)
python compare_uprights.py to_upright_*.npz          # their stiffness and frequencies
```

`vox_current` is whatever `build_cnc.py` draws now (`voxelize.py` rewrites it); `vox_previous` is the frame before these changes, which the optimizations and the "previous design" numbers above use. `VOX_P='{"beam_d": 160}' VOX_OUT=analysis/vox_test` on the voxelize command tries a parameter change.

## Chips and coolant

| Stage | What | Where in the model |
|---|---|---|
| Collect | Sealed base top; gutters between the uprights and the enclosure walls fall toward a trough across the front, which falls to one outlet | `BaseFrontCasting`, `BaseRearCasting` |
| Protect | Roll-up 0.1 mm stainless strips in front of and behind the table, a spiral spring cover on the rear screw, a drip shield over the beam rails, brush seals where the motors pass through the panels | `WayCovers`, `CoverRoller*`, `DripShield`, `BrushSeal*` |
| Drain | The trough drops straight down through the front wall to a cast-in brass nipple over the chip basket | `DrainOutlet` |
| Clean | Slide-out sump under the base, between four 6 in steel legs cast into the corners: a full-size 4 in stainless steam-table pan (~13 L, ~9 L working). Front to back: 1/3-size perforated pan as the chip basket, magnet bar, printed underflow baffle, 100 um screen in a printed frame, pump | `Sump*`, `Leg*` |
| Return | 24 V brushless submersible pump (switched by the controller's M8 output; no mains in the coolant), push-to-connect at the pump, hose under the base and up the back to a bulkhead in the back panel, along the printed cable chain, a loop down to a manifold on the spindle clamp, two Loc-Line nozzles | `ReturnHose`, `Bulkhead`, `CoolantLines`, `CoolantNozzle*` |

To empty or clean the sump, pop the hose off the pump and slide the pan out the front (set `sumpOut` in the `Params` sheet to see it pulled out). The legs raise the machine 117 mm and it still sits on four M12 leveling feet, now in the leg ends. Use a water-soluble coolant and keep it topped up; skim tramp oil off the clean end now and then.

**Cable chain.** One printed chain on top of the beam carries the coolant line, Z motor, spindle power and spindle water hoses to the carriage (17 links, 38 x 21 outer, 20 mm pitch, R38). Print it in PETG, not PLA. The model shows a simplified link; for the real parts, start from a proven printable parametric chain sized to about 30 x 15 inside and print three links to check the fit before printing the rest. The table and Z axes don't need chains: the table carries no cables and Z uses a short service loop.

**Parts that change shape with the axes.** The way-cover strips, the screw cover, the cable chain and the nozzle hose are drawn for the home position. The collision check and the renders rebuild them at every pose, but jogging `posX` / `posY` / `posZ` in the GUI won't redraw them; re-run the macro to see them elsewhere.

## Cost

**Checked against real listings (2026-10-09/10, the sourcing list).** The list is in three parts: **1, the frame** (castings, steel, rails, ballscrews, chip protection) about **$1,647** (including $150 of alumina grit for the beam and uprights, and about $200 for the bolt-together mold kit: two sheets of melamine, its hardware, a sheet of XPS foam and the PETG); **2, the electronics and spindle** about **$430** plus the spindle the builder picks ($250-375 for a 1.5 kW 110 V kit, $430-600 for the 2.2 kW kit with a 120 V step-up VFD); **3, the enclosure and flood coolant** about $650-700. The steel and aluminum are priced as remnants and the joint hardware from a hardware store; the remnant prices are estimates. The estimate below was made before that check and is too low; it assumed AliExpress-level prices. It's kept here to show what each cost-down change saves.

Rough estimates in USD at hobby-supplier prices (AliExpress/Amazon-type sources, late 2026). They're not quotes; check current listings. The cost-down column keeps the machine's capability: same travel, rigidity, spindle, closed-loop X/Y, flood coolant and enclosure.

| Item | Before | Cost-down | Change |
|---|---|---|---|
| Epoxy + aggregate | $240-315 | $200-255 | Hollow rear base, 8.5 % epoxy mix (well-graded aggregate + whiting fines), bulk kits |
| Cast-in steel, rods, dowels, fasteners | $90-140 | $90-140 | |
| Rail risers | $60-90 | $25-40 | Cold-rolled flat bar bedded in epoxy and replicated, instead of MIC6 strips |
| MIC6 table, carriage, Z plate | $120-180 | $120-180 | Remnant/drop plate is often much cheaper |
| Rails + blocks (table, Z) | $110-160 | $110-160 | Beam rails owned |
| Ballscrew kits (table, Z) | $70-100 | $70-100 | Beam kit owned |
| X/Y closed-loop motors + drivers | $170-220 | $140-180 | 2 Nm 23HS30-E1000: lower inductance, holds torque better at speed, 37 mm shorter |
| Brackets, couplers | $20-30 | $20-30 | One bracket owned |
| Spindle, VFD, clamp, cooling loop | $280-370 | $280-370 | |
| Electronics | $150-200 | $150-200 | |
| Enclosure | $220-300 | $145-210 | 3 mm ACM walls and roof; polycarbonate only in the door |
| Coolant system | $105-180 | $105-180 | |
| Way covers, seals | $65-105 | $65-105 | |
| Lubrication | $40-60 | $10-15 | Grease gun on the blocks' grease nipples instead of a central pump |
| Gaskets, inserts, feet | $30-45 | $30-45 | |
| Printed cable chain | $3 | $3 | PETG |
| **Total** | **~$1,780-2,500** | **~$1,560-2,210** | |

Bigger savings mean giving something up. Options, roughly in order of impact:

- **Phase it.** Build without the enclosure and flood coolant first (mist or air blast). That saves about $250-400 up front. The base still has the gutters and mounting inserts, so both can be added later.

- **Polyester instead of epoxy** for the castings: about half the resin cost, but it shrinks more, so you rely more on the epoxy-replicated mounting surfaces.
- **1.5 kW 110 V spindle** instead of the 2.2 kW you can upgrade to 240 V later: about $40-70 cheaper, less power for steel.
- **Open-loop motors** on X/Y: about $60-80 cheaper, but you lose stall and crash detection.

## Reference dimensions

| Component | Source | Notes |
|---|---|---|
| HGR15 rail, HGH15CA block | HIWIN Linear Guideway catalog G99TE23-2203, HG series (p.38) | Ball-groove positions are not published; estimated. |
| SFU1204 / SFU1605 nuts, screw end machining | TBI Motion SFU sheet L010974 (via Anaheim Automation) | |
| BK10/BK12, BF10/BF12 supports | TBI Motion sheets L011175 (BK), L011178 (BF) | BK locknut size estimated. BK12 counterbore depth (1.5) is as printed; worth checking. |
| DSG16H nut housing | makerstore.cc drawing TR-BS-DSG16-00 | |
| DSG12H nut housing (24 mm bore) | zyltech / vxb listings | Center height and holes estimated. |
| NEMA 23 motors | StepperOnline drawings 23HS45-4204D-E1000, 23HS30-5004D-E1000, 23HS30-3004S | |
| Motor bracket | No published drawing | Sized from kit listings: motor face ~47 mm from the BK. |
| D25 L30 coupler | Seller listings (5 N·m, 8x10 / 6.35x10 bores) | Hub depth estimated. |
| GDZ80-2.2 spindle | StepperOnline drawings GDZ80-2.2A/B | Some features scaled from the drawing (±1 mm). Body length varies by seller (213/230/240): measure yours. |
| 80 mm cast clamp | cnconline.co.za 80 mm spindle bracket drawing | 148 W x 80 D, 14 foot, bore center 50 off the foot. |
| ER20 nut | DIN 6499 / spindle drawing | Chinese spindles ship a 30 AF hex nut. |
| 2020 extrusion | Misumi HFS5-2020 catalog p.2-511 | Owned V-slot is OpenBuilds profile (6.25 throat). |

Before casting anything, measure the parts you actually have, especially the owned ANWOK rails, blocks and SFU1204 kit. Clone parts can differ from these catalogs by a millimeter or two.